

Take acceleration due to gravity as  $10\text{m/s}^2$  and ignore air resistance.

1. A body is projected horizontally from the top of a building 125 metres high with a speed of 10 metres/second. Find:
  - (a) the time taken to reach the ground.
  - (b) the horizontal distance travelled.
 Explain why the time taken to reach the ground is independent of the velocity of projection.
2. A cricket ball is hit with a velocity of  $12\frac{1}{2}\text{ m/s}$  at an angle with the horizontal whose tangent is  $\frac{4}{3}$ . Find:
  - (a) the greatest height reached,
  - (b) the time of flight,
  - (c) the horizontal distance travelled.
3. At what angle should a particle be projected with a velocity of 50 m/s to pass through a point on its path whose coordinates are (200, 25), distances being measured in metres?
4. A ball is projected so that its horizontal range is 45 metres and it passes through a point  $22\frac{1}{2}$  metres horizontally from and  $11\frac{1}{4}$  metres vertically above the point of projection. Find the angle of projection and the speed of projection.
5. A stone is projected from the edge of a cliff with a speed of 30 metres per second to hit an object 120 metres horizontally from the edge and 35 metres vertically below it. Find the angle of projection.
6. A ball thrown horizontally with speed  $V$  from a point  $H$  metres above the ground lands at horizontal distance  $D$  metres from the point of release. Find
  - (a)  $V$  if  $D = 3$  and  $H = 1\frac{1}{4}$ ,
  - (b)  $H$  if  $V = 10$  and  $D = 20$
7. A particle is projected from a point whose coordinates are (0, 0) with velocity of 20 m/s horizontally and 100 m/s vertically. Find when the projectile passes through the point whose horizontal distance from  $O$  is 120 metres and find also its vertical height then.
8. A particle is projected so that at any time  $t$ , its position  $(x, y)$  is given by  $x = 36t$ ,  $y = 15t - \frac{1}{2}gt^2$  where the components are measured in metres, and time in seconds. Find:
  - (a) the velocity of projection,
  - (b) the angle of projection,

- (c) the greatest height reached,
- (d) the range.

9. A particle is projected so that at any time  $t$  the equation of the trajectory  $x = 15t$ ,  $y = 20t - \frac{1}{2}gt^2$  where the components are measured in metres. Find:
  - (a) the point of projection;
  - (b) the speed of projection;
  - (c) the coordinates of the highest point of the path.
10. A shell is fired at 200 metres/second to strike a target 2 km away. At what angle should it be fired?
11. A plane travelling at 630 km/h releases a bomb from a height of 1000 metres.
  - (a) How long does it take to reach the ground?
  - (b) What horizontal distance does the bomb travel?
12. A particle is projected to just clear two walls of height 7 metres and distant 7 metres and 14 metres from the point of projection. Prove that if  $\alpha$  is the angle of projection, then  $\tan \alpha = \frac{3}{2}$ .  
 Prove that if the walls are  $h$  metres high and distant  $b$  metres and  $c$  metres from the point of projection, then
 
$$\tan \alpha = \frac{h(b+c)}{bc}$$
13. A particle is projected with speed of 25 m/s in the direction of a point  $P$  distant 24 m horizontally and 7 m vertically above the point of projection.
  - (a) Find the coordinate of the point on the trajectory directly below  $P$ .
  - (b) At the instant of projection of the particle, a second particle is dropped from  $P$ .  
 Prove that the two particles will collide.
14. A particle projected at an angle of  $15^\circ$  has a horizontal range of 80 metres.
  - (a) What is the velocity of projection?
  - (b) What would the range be if the angle were increased to  $45^\circ$  and the same velocity of projection?
15. A stone is projected horizontally from the top of a building 30 metres high with a velocity of 15 m/s. At the same instant another stone is projected from the foot of the building with a velocity of 30 m/s at an angle of  $60^\circ$  to the horizontal. The two stones move in the same vertical plane, Prove that they collide and, if the coordinates of the



foot of the building are  $(0, 0)$ , find the coordinates of the point of collision.

16. A golfer hits a ball from a point on a level fairway and 2 seconds later it strikes the fairway 50 m away. Find:
  - (a) the velocity and angle of projection
  - (b) maximum height of the ball above the fairway.
17. A stone is thrown so that it will hit a bird at the top of a pole. However, at the instant the stone is thrown, the bird flies away in a horizontal straight line at a speed of  $10 \text{ ms}^{-1}$ . The stone reaches a height double that of the pole and, in its descent, touches the bird. Find the horizontal component of the velocity of the stone.
18. A particle is projected with velocity  $20 \text{ m/s}$  to hit a target at a horizontal distance  $20 \text{ m}$  from the point of projection and at a vertical height of  $10 \text{ m}$ . Find two possible angles of projection.
19. A particle is projected so that at any time  $t$  the co-ordinates of its position relative to a fixed point  $O$  are given by the parametric equations  $x = 30t$  and  $y = 40t - \frac{1}{2}gt^2$  where the components are measured in metres and time in seconds. Find:
  - (a) the velocity of projection
  - (b) the angle of projection
  - (c) the Cartesian coordinates of the highest point of the path.
20. A particle projected from a point meets the horizontal plane through the point of projection

after travelling a horizontal distance  $a$ , and in the course of its trajectory attains a greatest height  $b$  above the point of projection. Find the horizontal and vertical components of the velocity at  $t = 0$  in terms of  $a$  and  $b$ . Show that when it has described a horizontal distance  $x$ , it has attained a height of  $\frac{4bx(a-x)}{a^2}$ .

21. A body is projected with a velocity whose horizontal and vertical components are  $6 \text{ m/s}$  and  $4 \text{ m/s}$  respectively. Find the horizontal range.
22. A batsman hits a cricket ball 'off his toes' towards a fieldsman who is  $65 \text{ m}$  away. The ball reaches a maximum height of  $4 \cdot 9 \text{ m}$  and the horizontal component of its velocity is  $28 \text{ m/s}$ . Find the constant speed with which the fieldsman must run forward. Starting at the instant the ball is hit, in order to catch the ball at a height of  $1 \cdot 3 \text{ m}$  above the ground. (Use  $g = 9 \cdot 8$ )
23. A football, kicked at  $16 \text{ m/s}$ , just passes over a cross-bar  $4 \text{ m}$  high and  $16 \text{ m}$  away. Show that, if  $\alpha$  is the angle of projection,
 
$$5 \tan^2 \alpha - 16 \tan \alpha + 9 = 0$$
24. Find the speed and direction of a particle which, when projected from a point  $15 \text{ m}$  above the horizontal ground, just clears the top of a wall  $26\frac{1}{4} \text{ m}$  high(max) and  $30 \text{ m}$  away.

Answer:

- |                                       |                                                |                                                  |
|---------------------------------------|------------------------------------------------|--------------------------------------------------|
| 1. (a) $5 \text{ s}$                  | (d) $108 \text{ m}$                            | (b) $5 \text{ m}$                                |
| (b) $50 \text{ m}$                    | 9. (a) $(0, 0)$                                | 17. $5(\sqrt{2} + 1) \text{ m/s}$                |
| 2. (a) $5 \text{ m}$                  | (b) $25 \text{ m/s}$                           | 18. $45^\circ, 71^\circ 34'$                     |
| (b) $2 \text{ s}$                     | (c) $(30, 20)$                                 | 19. (a) $50 \text{ m/s}$                         |
| (c) $15 \text{ m}$                    | 10. $15^\circ \text{ or } 75^\circ$            | (b) $53^\circ 8'$                                |
| 3. $36^\circ 52', 60^\circ 15'$       | 11. (a) $10\sqrt{2} \text{ s}$                 | (c) $(120, 80)$                                  |
| 4. $45^\circ, 15\sqrt{2} \text{ m/s}$ | (b) $\frac{7\sqrt{2}}{4} \text{ km}$           | 20. $\frac{a}{2}\sqrt{\frac{g}{2b}}, \sqrt{2gb}$ |
| 5. $36^\circ 52'$                     | 12. No answer                                  | 21. $4 \cdot 8 \text{ m}$                        |
| 6. (a) $6 \text{ m/s}$                | 13. (a) $(24, 2)$                              | 22. $7 \text{ m/s}$                              |
| (b) $20 \text{ m}$                    | 14. (a) $40 \text{ m/s}$                       | 23. No answer                                    |
| 7. $6 \text{ s}; 420 \text{ m}$       | (b) $160 \text{ m}$                            | 24. $25 \text{ m/s}, 36^\circ 52'$               |
| 8. (a) $39 \text{ m/s}$               | 15. $(10\sqrt{3}, 23\frac{1}{3})$              |                                                  |
| (b) $22^\circ 37'$                    | 16. (a) $5\sqrt{29} \text{ m/s}, 21^\circ 48'$ |                                                  |
| (c) $11\frac{1}{4} \text{ m}$         |                                                |                                                  |